Research Intern Candidate | Robotics, Trustworthy Motion Planning | PhD @ Duke ECE

RUIYANG WANG

Durham, NC, 27701 • (984)-259-5539 • ruiyang.wang@duke.edu • https://ruiyangwangw.github.io

Summary

Applied ML and Robotics researcher experienced in building scalable robot learning and planning systems with **PyTorch** and **ROS2**. Skilled in **multi-robot coordination**, **trust estimation**, and **language conditioned reasoning**. Hands-on with **Dockerized simulation**, **modular pipeline design**, and **real-world robotic deployment**.

Skills

Programming & Tools: Python, C++, Julia, MATLAB, ROS2, Linux/Unix, Git, Docker
Tools & Cloud: AWS S3/EC2 , MLflow, Weights & Biases, OpenAI API
ML/AI Frameworks: PyTorch, TensorFlow, scikit-learn
Robotics & Simulation: ROS2, LCM, PyBullet, OpenAI Gym, Nvidia Isaac Lab

Education

Duke University - Durham, NC *August 2024 - April 2028 (Expected)*
Doctor of Philosophy (Ph.D.): Electrical and Computer Engineering

University of Michigan, Ann Arbor - Ann Arbor, MI *August 2018 - December 2023*
Master of Science: Robotics
Bachelor of Science: Mechanical Engineering

Applied ML/Robotics Projects

Neural-Symbolic Trust Graphs for Adversarial Agent Detection *September 2025 – Present*
PyTorch, Graph Attention Networks, Agent Trust

- Modeled robot and object interactions as ego-centric **heterogeneous graphs** with **symbolic** behavioral **edges** encoding co-detection, missed detections, and relational context.
- Developed a **GAT-based evidence** extractor to induce implicit **cross-validation** among robots based on structural consistency of their detections.
- Integrated per-agent GNN outputs into **Beta** distribution **trust** updates, allowing trust estimates to evolve based on detection agreement patterns **across time**.

Large Language Model Multi-robot Coordinated Exploration and Search *January 2025 - September 2025*
PyTorch, ROS2, OpenAI API, Unitree Go2

- Built **LLM-MCoX**, a large-scale multi-robot coordination framework integrating LiDAR frontier clustering, doorway detection, and LLM reasoning.
- Enabled **semantic guidance** from natural-language priors (e.g., "object likely in northeast corridor") to optimize exploration and search tasks.
- Validated in 100+ simulated and real-world trials with Unitree Go2 and X500 drone; achieved 22% faster exploration and **50% higher search efficiency** than baseline frontier-based methods. **[P1]**

* denotes for equal contributions

Neural-Symbolic Deadlock Resolution in Multi-robot Systems *August 2024 - December 2024*
PyTorch, PyBullet, Neural Logic Machines

- Built a **Neuro-Symbolic planner** combining Neural Logic Machines for post-deadlock resolution in multi-robot systems.
- Trained models on **small 2–5 robot scenarios**, generalizing to **10+ robot** configurations in cluttered layouts. Ensured post-occurrence deadlock resolution in multi-robot coordination, where existing methods often resulted in incomplete task execution. **[P2]**

Compatible Constraint Selection in Receding-Horizon Control *December 2022 - December 2023*
Optimization, MPC, MATLAB, Python

- Implemented heuristics for **maximal feasible subset selection** in nonlinear MPC using Lagrange multipliers to identify conflicting soft constraints.
- Achieved **10× lower computation cost** than exhaustive and reachability-based methods with **near-optimal** task performance under obstacle and disturbance scenarios. **[P3]**

Selected Publications

- **[P1] Ruiyang Wang**, Hao-lun Hsu, David Hunt, Shaocheng Luo, Jiwoo Kim and Miroslav Pajic, "LLM-MCoX: Large Language Model Multi-robot Coordinated Exploration and Search", Submitted to *International Conference on Robotics & Automation (ICRA)* 2026.
  Video demonstrations available from
  https://ruiyangwangw.github.io/assets/ICRA_2026_LLM-MCoX_video.mp4
- **[P2] Ruiyang Wang**, Bowen He, Miroslav Pajic, "Neural Symbolic Deadlock Resolution in Multi-robot Systems", Accepted to *Learning for Dynamics & Control (L4DC)* 2025.
  Available from https://proceedings.mlr.press/v283/wang25f.html
- **[P3] Ruiyang Wang***, Hardik Parwana*, Dimitra Panagou, "Algorithms for Finding Compatible Constraints in Receding-Horizon Control of Dynamical Systems", Accepted to *American Control Conference (ACC)* 2024.
  Available from https://ieeexplore.ieee.org/abstract/document/10644243

Selected Course Projects

Deep Reinforcement Learning & Policy Optimization

- Developed a **Proximal Policy Optimization (PPO)** agent from scratch in **PyTorch** using an Actor-Critic architecture to solve both discrete (*CartPole*, *Acrobot*) and continuous (*LunarLander*) control environments.
- Conducted ablation studies comparing standard PPO against **GRPO**, demonstrating that learned value function baselines significantly reduce training variance and oscillation.

Retrieval-Augmented Generation (RAG) & BERT Optimization

- Engineered a semantic search pipeline using **Sentence-BERT** and **Cross-Encoders** to retrieve top-k relevant NeurIPS abstracts. Integrated retrieved context with **GPT-2** to generate factually grounded answers, significantly reducing hallucinations compared to prompt-only generation.
- Fine-tuned **BERT** for sentiment classification on a **5k-sample subset** of the Yelp dataset (560k total). Achieved **~94% validation accuracy** with full fine-tuning, significantly outperforming the **66% accuracy** of the linear probing baseline.